

Scenario 3: Hotel Revenue Protection & Cancellation Analysis

1. The Business Story

Role: Revenue Analyst for an international hotel group (managing both a City Hotel and a Resort Hotel).

Business Problem: Over 35% of all reservations end up canceled before check-in. This causes empty rooms during peak seasons and massive revenue loss. The General Manager needs to know:

- *What is the financial impact of cancellations?*
- *Which customer segments and booking channels have the highest cancellation rates?*
- *How far in advance (`lead_time`) do cancellations happen, and can we identify high-risk bookings early?*

Dataset

- **Dataset Link:** [Kaggle - Hotel Booking Demand \(by Jesse Mostipak\)](#)
- (Download `hotel_bookings.csv`)*

Key Columns:

`hotel`, `is_canceled`, `lead_time`, `arrival_date_year`, `arrival_date_month`, `stays_in_weekend_nights`, `stays_in_week_nights`, `adults`, `children`, `country`, `market_segment`, `deposit_type`, `customer_type`, `adr` (Average Daily Rate).

What Students Do in Python (`pandas` & `numpy`)

1. Data Cleaning:

- Filter out "ghost bookings" where total guests equals 0: `adults + children + babies == 0`.
- Handle missing values: Fill missing `country` with "Unknown" and impute missing `children` with 0.
- Identify and remove `adr` anomalies using `numpy` (e.g., negative room prices or extreme outliers like $> 5,000/\text{night}$).

2. Feature Engineering:

- **Total Stay:** `total_stay = stays_in_weekend_nights + stays_in_week_nights`
- **Estimated Booking Value:** `total_cost = adr * total_stay`

- **Revenue Lost Flag:** Use `np.where()` to calculate `revenue_lost` (if `is_canceled == 1`, capture `total_cost`, else \$0).
- **Lead Time Buckets:** Use `pd.cut()` or `np.select()` to group `lead_time` into buckets:
 - *Last Minute* (0–7 days)
 - *Short Notice* (8–30 days)
 - *Medium* (31–90 days)
 - *Far Advance* (> 90 days)
- **Family/Group Indicator:** Flag whether the booking includes children/babies using `np.where()`.

3. **Export:** Export clean data to `clean_hotel_bookings.csv`.



What Students Build in Tableau

- **Revenue Impact KPIs:** Big Number Cards for *Overall Cancellation Rate (%)*, *Total Realized Revenue*, and *Total Lost Revenue from Cancellations*.
 - **City vs. Resort Comparison:** Side-by-side bar/line chart tracking cancellation rates and ADR across months.
 - **Lead Time vs. Cancellation Risk:** Heatmap or scatter plot showing cancellation likelihood across different `lead_time` buckets and market segments (e.g., *Online Travel Agents* vs. *Direct Bookings*).
 - **Geographical Distribution:** World map highlighting which guest countries bring in the most revenue vs. which have the highest cancellation rates.
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Short Email Template to Send to Students

Subject: New Portfolio Project: Hotel Cancellation & Revenue Analysis (Python + Tableau)

Hi everyone,

For our next project, you will step into the role of a **Revenue Analyst** for a hotel chain looking to combat empty rooms and revenue leakage.



The Business Challenge:

Over a third of reservations get canceled before arrival. Management needs an end-to-end data pipeline and dashboard to identify **which bookings are at high risk of cancellation** and **how much revenue is being lost**.

Dataset:

Download `hotel_bookings.csv` from Kaggle:

 [Hotel Booking Demand Dataset](#)

Key Tasks:

1. Python (`pandas` & `numpy`):

- Clean messy data: handle missing values in `country` and `children`, and remove ADR outliers (e.g., negative room rates).
- Feature engineering: Calculate `total_stay`, total booking value ($\text{adr} * \text{total_stay}$), and categorize `lead_time` into buckets (*Last Minute, Short Notice, Far Advance*).
- Export the clean dataset to `.csv`.

2. Tableau Dashboard:

- Build an interactive executive dashboard covering:
 - Revenue Loss & Cancellation Rate KPIs.
 - Monthly cancellation trends: **City Hotel** vs. **Resort Hotel**.
 - Cancellation risk across booking lead times and market segments.
- Publish your dashboard to **Tableau Public**.

3. GitHub Requirements:

- Push your clean, commented Jupyter Notebook (`.ipynb`).
- Write a professional `README.md` with:
 - A summary of the business problem.
 - 3 data-driven recommendations (e.g., *"Require deposits on bookings made >90 days in advance"*).
 - An image preview + link to your live Tableau dashboard.

 **Deadline:** [Insert Date]

This is a classic revenue management use case that looks great to analytics hiring managers. Start early and let me know if you hit any roadblocks!

Best regards,
Kamran Saberifard